MODULE 03 · SECTION 1 - FUNDAMENTALS OF GEN AI

How machines learn: supervised, unsupervised, reinforcement

Three ways a model learns, and the words in a question that give each one away.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 3 of 33

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You rarely need to know the algorithm. Look at **what the data looks like**, and the learning type follows.

Three ways to learn

APPROACH	LEARNS FROM	PRODUCES	THINK OF
Supervised	Labelled examples	A predicted label or number	Studying with an answer key
Unsupervised	Unlabelled data	Groups, patterns, oddities	Sorting a jar of buttons
Reinforcement	Rewards and penalties	A strategy for acting	Training a dog with treats

THE TIEBREAKER

Labels present, labels absent, or rewards for actions. Three different starting points, three different types.

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Supervised: learning from the right answers

- **Labelled data:** every example is paired with the correct answer.
- **The model adjusts** itself to make fewer mistakes against those answers.
- **Examples:** spam filtering, sentiment, predicting flight times, pricing.
- **Strength:** accurate, measurable predictions for a known question.
- **Cost:** someone has to create the labels, and that takes time and money.

> ON THE EXAM
>
> Historical records with known outcomes are labelled data. Past loans marked repaid or defaulted are a textbook example.

Classification or regression?

CLASSIFICATION

Predicts a category

- Spam or not spam
- Positive, negative or neutral
- Will this customer churn?
- Which of 40 product types?

REGRESSION

Predicts a quantity

- How many days will delivery take?
- What price will this sell for?
- How many units next month?
- How long until the ticket closes?

THE TIEBREAKER

Both are supervised. A category means classification; a number on a scale means regression.

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Unsupervised: finding structure nobody labelled

- **No labels.** The model learns the data's own structure without guidance.
- **Clustering:** natural groups, such as customer segments.
- **Anomaly detection:** records that do not fit the usual pattern.
- **Strength:** reveals patterns you did not know to look for.
- **Limit:** it finds groups but cannot name them. People interpret the results.

> ON THE EXAM
>
> Words like discover, natural groupings, unknown segments, or no labels yet all point to unsupervised learning.

Reinforcement: learning by trial and error

- **An agent** acts in an **environment**, one action at a time.
- **Rewards and penalties** tell it which actions led somewhere good.
- **Over many attempts** it learns a strategy that earns the most reward.
- **Uses:** games, robotics, routing, and adapting recommendations.
- **Also** how chat models are taught to prefer helpful answers, using human feedback.

> ON THE EXAM
>
> Scores, rewards, penalties, trial and error, or learning by doing: that is reinforcement learning.

The words that give each type away

WHEN THE SCENARIO SAYS	IT IS
labelled, tagged, historical outcomes, known answers	Supervised
predict a category, approve or reject, spam or not	Supervised classification
predict an amount, a price, a time, a count	Supervised regression
no labels, discover, natural groupings, segments	Unsupervised clustering
unusual, outliers, does not fit the pattern	Unsupervised anomaly detection
reward, penalty, score, trial and error	Reinforcement

THE TIEBREAKER

Underline the data words first. They settle most of these questions before you reach the options.

Deep learning is **not a fourth type**. It is a technique that can be supervised, unsupervised or reinforcement.

How a foundation model uses all three ideas

- **Pre-training** on vast unlabelled text: the text itself supplies the next word to predict.
- **Supervised tuning** on labelled examples of good responses.
- **Reinforcement** from human feedback, so preferred answers are rewarded.
- **Agent Platform** offers supervised tuning and preference tuning for Gemini.

> REMEMBER
>
> You will not be asked to build any of this. It explains why one model can do so much after so many stages.

Wrenfield's ideas, by learning type

WRENFIELD IDEA	THE DATA	LEARNING TYPE
Cardholder will miss payments?	Accounts with known outcomes	Supervised classification
Days a delivery will take?	Past deliveries, actual days	Supervised regression
What kinds of shopper exist?	Purchases, no segments	Unsupervised clustering
Which refunds look unusual?	Refunds, no fraud labels	Unsupervised anomalies
Best route for the robot?	Rewards for fast, safe picks	Reinforcement

THE TAKEAWAY

One company needs all three. The starting data decides which.

> Wrenfield Pay holds five years of store-card accounts. Each one is marked with whether the customer kept up repayments or fell behind.
>
> The credit team wants a model that estimates whether a new cardholder is likely to fall behind.

Which type of machine learning fits?

A Unsupervised learning

B Supervised learning

C Reinforcement learning

D Deep learning

◆ PAUSE AND DECIDE

Which type of machine learning fits?

A Unsupervised learning

The outcomes are known, so there is no need to discover groups.

B Supervised learning

Historical accounts with known outcomes are labelled data.

C Reinforcement learning

No agent, no actions, no rewards.

D Deep learning

A technique, not a learning type.

THE SIGNAL

Each account is marked with its outcome. Marked outcomes are labels.

> Wrenfield has 18 months of contact-centre calls, transcribed to text. Nobody ever recorded why each customer called.
>
> The operations director wants to find the main reasons people call, including reasons nobody has thought of.

Which approach fits best?

A Supervised classification into the five reasons managers expect

B Reinforcement learning that rewards shorter calls

C Unsupervised clustering of the transcripts

D Regression to predict call length

◆ PAUSE AND DECIDE

Which approach fits best?

A Supervised classification into the five reasons managers expect

No labels exist, and fixed categories cannot reveal new reasons.

B Reinforcement learning that rewards shorter calls

Optimises behaviour. It does not explain why people call.

C Unsupervised clustering of the transcripts

Groups similar calls with no labels, revealing unexpected themes.

D Regression to predict call length

Predicts a number. It does not reveal reasons.

THE SIGNAL

No labels, plus reasons nobody has thought of. Discovery means unsupervised.

> A logistics firm is training a warehouse robot to pick orders. The robot earns points for each fast, correct pick and loses points for collisions or wrong items.
>
> Over thousands of practice runs, its routes improve.

What type of machine learning is this?

A Supervised learning

B Unsupervised learning

C Deep learning

D Reinforcement learning

◆ PAUSE AND DECIDE

What type of machine learning is this?

A Supervised learning

No labelled examples of correct routes are provided.

B Unsupervised learning

It is not finding groups. It is learning to act.

C Deep learning

Might power the robot, but it is a technique, not the learning type.

D Reinforcement learning

Points gained and lost for actions, over many attempts.

THE SIGNAL

Earns points, loses points, improves over practice runs: rewards for actions.

> Wrenfield promises delivery dates for sofas, and customers complain when they are wrong. The company has records of 200,000 past deliveries, each with the actual number of days taken.
>
> It wants a model that predicts the number of days for each new order.

Which kind of task is this?

A Supervised regression

B Supervised classification

C Unsupervised clustering

D Reinforcement learning

◆ PAUSE AND DECIDE

Which kind of task is this?

A Supervised regression

Known past outcomes, and the answer is a number of days.

B Supervised classification

Right family, but the output is a quantity, not a category.

C Unsupervised clustering

The outcomes are known, and no groups are wanted.

D Reinforcement learning

No agent learning from rewards.

THE SIGNAL

Labelled history plus a numeric answer: regression.

> A hospital pharmacy logs every dispensing event. It has never recorded a confirmed case of misuse, so there are no examples to learn from.
>
> Managers want unusual dispensing patterns flagged for a pharmacist to review.

Which approach fits best?

A Supervised classification trained on past misuse cases

B Reinforcement learning that rewards correct dispensing

C Unsupervised anomaly detection

D A generative model that writes a daily summary

◆ PAUSE AND DECIDE

Which approach fits best?

A Supervised classification trained on past misuse cases

There are no past cases to train on.

B Reinforcement learning that rewards correct dispensing

Nothing is acting or being rewarded.

C Unsupervised anomaly detection

Flags events that do not fit the usual pattern, with no labels.

D A generative model that writes a daily summary

Summarising does not find the unusual events.

THE SIGNAL

No labelled examples, and the goal is spotting the unusual. That is anomaly detection.

> Wrenfield has 60,000 product photos, each already tagged with one of 40 product types. A model will learn to tag new photos.
>
> One team member says this is deep learning, so it cannot be supervised learning.

Which statement is most accurate?

A Deep learning and supervised learning are different types, so it is one or the other

B It is supervised learning, and deep learning is the technique likely used

C It is unsupervised, because images contain no labels

D It is reinforcement learning, because the model improves over time

◆ PAUSE AND DECIDE

Which statement is most accurate?

A Deep learning and supervised learning are different types, so it is one or the other

They are not rivals. Deep learning is a technique used within the types.

B It is supervised learning, and deep learning is the technique likely used

Tagged photos are labels; neural networks do the learning.

C It is unsupervised, because images contain no labels

These photos are already tagged with product types.

D It is reinforcement learning, because the model improves over time

Every type improves with training. There are no rewards for actions.

THE SIGNAL

Labels decide the type. Deep learning describes the machinery, not the type.

Module 3 in five lines

- Supervised learns from labelled examples; known historical outcomes count as labels.
- Classification predicts a category; regression predicts a number. Both are supervised.
- Unsupervised finds structure in unlabelled data: clusters and anomalies.
- Reinforcement learns a strategy from rewards and penalties, by trial and error.
- Deep learning and generative AI are not learning types. Choose from the three.

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